Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. MEANINGFUL: (A) solitary (B) theoretical (C) insignificant (D) incomplete (E) familiar

2. ADAPTABLE: (A) selfish (B) unwholesome (C) inflexible (D) irritable (E) suspicious

3. DISBAND: (A) obey (B) reverse (C) ascend (D) assemble (E) confess

4. EVAPORATE: (A) provide (B) sparkle (C) cleanse (D) devastate (E) condense

5. OBSTINATE: (A) intermittent (B) yielding (C) uncertain (D) careless (E) despairing

6. COMMUNAL: (A) merciful (B) partial (C) neutral (D) liable (E) individual

7. EON: (A) short time (B) polite refusal (C) valuable article (D) vague impression (E) abrupt explanation

8. LUSH: (A) incisive (B) unvarying (C) barren (D) miniscule (E) solid

9. POLARIZATION: (A) incineration (B) consolidation (C) eradication (D) regeneration (E) validation

10. FLAUNT: (A) drift (B) devote (C) interrupt (D) hide (E) sicken

11. AFFINITY: (A) coarseness (B) limit (C) repulsion (D) angularity (E) uniformity

12. MITIGATE: (A) hesitate (B) receive (C) identify (D) review (E) aggravate

13. EVANESCENT: (A) descriptive (B) causal (C) detrimental (D) protective (E) permanent

14. EXPUNGE: (A) degrade (B) relieve (C) command (D) press (E) record

15. HEDONIST: (A) atheist (B) heretic (C) ascetic (D) renegade (E) novice

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. She was ---- success and eager to create a name for herself.

(A) deprecatory toward
(B) impatient for
(C) querulous about
(D) sated by
(E) impassive to

17. Since what constitutes a proper diet remains a ---- subject on which few experts agree, additional research is ----.

(A) controversial. .required
(B) neglected. .deplored
(C) popular. .unexpected
(D) favored. .prohibited
(E) complex. .redundant

18. Although the Civil War ---- much of the South's railroad network, it ---- the growth of railroads in the North and West.

(A) eliminated. .decreased
(B) accounted for. .encouraged
(C) displaced. .dismantled
(D) destroyed. .stimulated
(E) depended on. .established

19. Diversity is essential in spite of the fact that it ---- universal acceptance of a single doctrine.

(A) underlies (B) reserves (C) presupposes (D) entails (E) precludes

20. He makes his attack with a massive display of scholarship but no great ----; all too often the ---- rather than the rapier is his chosen weapon.

(A) pedantry. .bomb (B) elaboration. .ax (C) emphasis. .pen (D) discrimination. .dart (E) subtlety. .bludgeon

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

For the artist in the Black Art movement there is no separation between art and the oppressive conditions of society. Art is meant to be removed from the museums and displayed on tenement and playground walls, in schools and recreational centers, in storefront galleries, and on corner lots. Much of the work is collective; therefore, the personality of the artist is of secondary importance to the message or the content. The paintings are deliberately naïve, the forms simple and bold, and there is little subtlety in color relationships. Many movement murals have been painted by neighborhood children or young adults as storefront art center projects; others have been executed by college students under the direction of an artist-in-residence. Frequently, the murals are designed to teach the young about Blacks who have contributed to the growth of the United States. In institutions attended by Black children the murals thus become history lessons, and the Black leaders portrayed, men and women to be emulated.

The American flag is a recurrent theme in the paintings of many movement artists. In this respect and in others, their work resembles that of the Pop artists of the fifties and sixties. Pop Art attempted to erase the boundaries between "fine art" and the messages communicated by the mass media. Its images are large and bold, and there is some suggestion, at least in spirit, of a type of painting practiced by early colonial artists. Pop embraced the objects constituting the most public, everyday facts of modern life and used them in such a way as to criticize that style of life. There is one great difference, however, between Pop Art and the Black Art movement. While the Pop artists tended to objectify their images, stripping them of their original significance, the movement artists are deeply committed to the images and forms of their work.

A group of Chicago movement artists issued a statement about their goals. In the statement, the members of the group declare that because the Black artist and the creative potential of the Black experience have been excluded from the full spectrum of American arts in the past, they want to provide a new form and context for Black artists in which they can work out problems and pursue aims unhampered and uninhibited by the art establishment. The work of the group members is part of the movement effort to strip art of overrefinement and overemphasis on traditional museums; it supports the effort to create a kind of Black museum in the inner city. In essence these artists are saying to their people, "Our art is yours, you are us, and we are you."

21. All of the following statements about movement murals are supported by the passage EXCEPT:
    (A) They are often communal efforts.
    (B) They are seldom displayed in museums.
    (C) They are usually abstract.
    (D) They may have a historical subject.
    (E) They are concerned with standards and values.

22. The author discusses Pop Art primarily in order to
    (A) discredit the importance of Pop Art
    (B) imply that all art shares the same ideals
    (C) propose that movement artists must refer to earlier trends in art
    (D) predict that movement art will gain greater acclaim
    (E) illustrate some distinctive qualities of movement art

23. According to the passage, movement and Pop artists are similar insofar as both
    (A) refuse to exhibit their work in the traditional manner
    (B) employ a vivid communicative style
    (C) preserve their subjects' initial meaning
    (D) propose to supervise social projects
    (E) consider art to be a group enterprise

24. It can be inferred from the passage that movement artists of the Chicago group believe that museums are
    (A) aesthetically inadequate to display their work
    (B) representative of exclusive social practices
    (C) useful only as a means of acquiring recognition
    (D) potentially useful for educational purposes
    (E) responsible for informing Blacks of their past

25. Which of the following most accurately states the artists' position as implied by the last sentence of the passage?
    (A) All art is equally worthy of praise.
    (B) All innovative art is attractive to the general public.
    (C) Art is an activity that supersedes all others.
    (D) Art and society influence each other.
    (E) Socially aware artists must disguise their real objectives.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

GO ON TO THE NEXT PAGE

In 1834 the chemist Jöns Jakob Berzelius analyzed a peculiar meteorite that had fallen near Alais in France. He found that it contained carbonaceous material and he wondered: Does it contain humus or traces of other organic compounds? Does it possibly indicate the presence of organisms on extraterrestrial bodies? Studying the meteorite, Berzelius noted its resemblance to hardened clay. He established the presence of carbon compounds, but the state of organic chemistry in 1834 precluded further analysis. On the basis of his findings Berzelius decided that the presence of "organic" compounds in the meteorite did not "justify the conclusion that organisms existed in its original locality."

Meteorites in which organic compounds have been identified are called carbonaceous chondrites. These are a subgroup of the large family of stony meteorites known as chondrites because they contain small round bodies—chondrules—of the magnesium-iron silicates olivine and pyroxene. Sooty, black, friable objects, they do not look at all like other meteorites. They are characterized by the presence in small quantities (usually less than 1 per cent of the total weight of the sample) of carbonaceous material other than free carbon. Most of the carbon is neither volatile nor soluble, and it is present in fine-grained sooty material that defies precise analysis even with modern techniques. The smallness of carbonaceous chondrites (the largest on record is about the size of a human head: other stony meteorites weigh as much as a ton) is a reflection of the great amount of ablation, or vaporization, they suffer in their fiery passage through the atmosphere and of the readiness with which they break up. The fireballs accompanying their fall are extremely brilliant even when only a few small stones are recovered.

From about 1890 to 1950, little was published on the carbonaceous chondrites and nothing at all on the organic compounds in them. Then a renewal of interest in meteorites once again focused attention on these remarkable objects. The first modern investigation of the organic compounds was that of George G. Mueller, who analyzed the carbonaceous material in the Cold Bokkeveld meteorite. With organic solvents he extracted 1.1 per cent of the sample as a resinous material consisting of organic compounds mixed with free sulphur. A solution of the material in benzene did not rotate the plane of polarized light, which indicated that the material contained no optically active compounds. This finding was significant because optically active compounds are universally present in living organisms. The optical activity of hydrocarbons that have originated in biological material is low, however, and the level of activity could have been less than the sensitivity of Mueller's instrument.

26. The tone of the author's discussion of the work of Berzelius can best be described as

(A) skeptical (B) impersonal (C) apologetic (D) enthusiastic (E) indignant

27. The author's mention of "the state of organic chemistry in 1834" (line 10) indicates that

(A) Berzelius was unable to pursue his analysis further
(B) Berzelius jumped to an unwarranted conclusion
(C) Berzelius should have carried his experiments further
(D) Berzelius' findings should be disregarded
(E) Berzelius' findings were accurate

28. The author mentions the fireballs accompanying the fall of carbonaceous chondrites in order to emphasize

(A) the great amount of carbon in their makeup
(B) the ease with which they can be identified
(C) the improbability of their containing organic substances
(D) the amount of ablation they undergo in the atmosphere
(E) their similarity to other meteorites

29. Mueller's determination that his sample contained no optically active compounds suggests that

(A) the sample was too old to allow an accurate determination
(B) the experiment had not been performed properly
(C) the carbon compounds in the sample probably had not been present in living organisms
(D) living organisms in his sample probably did not contain such compounds
(E) there were no carbon compounds in the sample with which he was working

30. According to the author, one possible reason for Mueller's failure to detect optically active compounds in his sample is that

(A) such compounds cannot be analyzed by any known method
(B) their activity is so slight as to be very difficult to detect
(C) the compounds being studied were insoluble
(D) the presence of sulfur in the meteorite interfered with the analysis
(E) the meteorite being studied was not a carbonaceous chondrite

GO ON TO THE NEXT PAGE

1

Select the word or set of words that best completes each of the following sentences.

31. Her sense of time was oddly ----: ten minutes sometimes seemed like an hour.

(A) credible (B) distorted
(C) idealized (D) consistent (E) formidable

32. Narrowing the eyes when startled by a sudden noise or a noxious stimulus is ---- response, serving to reduce the exposure of sense organs to injury.

(A) an imaginative (B) a protective
(C) a corrective (D) a therapeutic
(E) an impaired

33. As a people, the Aztecs were full of contradictions: ---- and yet prizing humility, ---- and yet obliging and kind.

(A) arrogant. .benevolent
(B) pious. .gallant
(C) provincial. .sullen
(D) tolerant. .competitive
(E) proud. .implacable

34. His brilliant prose style is sometimes ---- by his lapses into ---- reflection and pretentious language.

(A) achieved. .facile
(B) extended. .whimsical
(C) disrupted. .understated
(D) anticipated. .personal
(E) marred. .banal

35. The study of crowds and groups is more than ---- in an age characterized by interdependence and ---- action.

(A) academic. .collective
(B) difficult. .intellectual
(C) vigorous. .assertive
(D) erudite. .formal
(E) general. .specific

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

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36. GOGGLES : EYES :: (A) thumbs : fingers
(B) noises : ears (C) stilts : legs
(D) gloves : hands (E) dimples : cheeks

37. PARAGRAPH : SENTENCES :: (A) cover : pages
(B) book : chapters (C) grammar : errors
(D) directory : graphs (E) summary : comments

38. PAW : BEAR :: (A) tail : monkey (B) gill : fish
(C) wing : bird (D) cub : lion (E) hoof : zebra

39. IMPLY : PROFESS :: (A) propose : negate
(B) vacillate : pause (C) hint : announce
(D) contemplate : deliberate (E) invite : accept

40. CALLOUS : SENSITIVITY ::
(A) bold : timidity
(B) faultless : perfection
(C) melodramatic : agitation
(D) discriminating : brevity
(E) detrimental : effectiveness

41. FLOE : ICE :: (A) dune : sand (B) bridge : river
(C) wake : vessel (D) desert : water
(E) blight : plant

42. AMORPHOUS : SHAPE :: (A) aimless : purpose
(B) infamous : wealth (C) ambiguous : obscurity
(D) stoic : philosophy (E) experimental : idea

43. SOPORIFIC : SLEEP :: (A) antidote : disease
(B) narcotic : stupor (C) sedative : tension
(D) tonic : immunity (E) liniment : vigor

44. DABBLE : SUPERFICIAL ::
(A) explain : confused (B) repudiate : angry
(C) surmise : impartial (D) deter : intelligent
(E) invent : resourceful

45. JINGOIST : NATION :: (A) fool : ridicule
(B) manager : personnel (C) braggart : self
(D) expert : contestant (E) artist : portrait

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

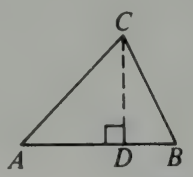
Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

Day	Number Absent
Mon.	7
Tues.	3
Wed.	2
Thurs.	5
Fri.	8

1. What is the average daily absentee rate for the period shown above?

   (A) 2 (B) 3 (C) 5 (D) 7 (E) 25

---

2. A photograph album contains exactly 25 pages numbered consecutively beginning with 1. There are exactly 4 photographs on each even-numbered page and exactly 5 on each odd-numbered page. What is the total number of photographs in the album?

   (A) 108 (B) 112 (C) 113
   (D) 117 (E) 118

3. If $x = -1$, of the following, which is greatest?

   (A) x^2 (B) x^3 (C) $\dfrac{1}{x}$
   (D) $1 + x$ (E) $1 - x$

4. If $v = 3w$, which of the following is equal to $12w - 4v$?

   (A) 0 (B) w (C) $2\frac{2}{3}w$ (D) $8w$ (E) $32v$

$$\begin{array}{r} P \\ \times\ Q \\ \hline 15 \end{array} \qquad \begin{array}{r} S \\ \times\ Q \\ \hline 21 \end{array}$$

5. In the multiplication problems above, if P, Q, and S are single digits, then $P \times S =$

   (A) 6
   (B) 9
   (C) 18
   (D) 35
   (E) 105

GO ON TO THE NEXT PAGE

6. A movie cartoon consists of 4,500 drawings. The animation artist did the first drawing, the sixth, and every fifth drawing thereafter. Of the following drawings in the sequence, which one did the artist produce?

(A) 31st (B) 24th (C) 20th
(D) 15th (E) 14th

7. $\frac{1}{3} \div \left(\frac{1}{3} \div \frac{1}{3} \right) =$

(A) $\frac{1}{9}$ (B) $\frac{1}{3}$ (C) 1 (D) 3 (E) 9

8. For what value of x does a triangle with sides $x + 2$, $3x + 1$, and $7 - 4x$ have a perimeter equal to 11 ?

(A) 0 (B) $\frac{1}{4}$ (C) 1 (D) 7
(E) For no value of x

9. If one week a merchant's income is \$160 greater than his weekly rent and the following week his income is \$80 less than the rent, what is the difference between the incomes for the two weeks?

(A) \$40
(B) \$80
(C) \$160
(D) \$240
(E) It cannot be determined from the information given.

10. The expression "x squide y" is defined as follows: first, square x and then divide the result by the least integer greater than y that will produce an integer solution. The quotient obtained is x squide y. If $x = 12$ and $y = 5$, what is x squide y ?

(A) 2
(B) 12
(C) 24
(D) 29
(E) 36

11. Which of the following CANNOT have more than 2 points in common?

(A) A circle and a triangle
(B) A circle and a rectangle
(C) A line and a circle
(D) Two triangles
(E) A line and a triangle

12. If p, q, and r are the degree measures of the three angles of an isosceles triangle and $p = 100$, then $p - q =$

(A) 20 (B) 40 (C) 50 (D) 60 (E) 80

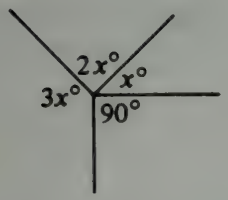
13. If $6x - 1$ is an odd integer, which of the following represents the next greater odd integer?

(A) $7x + 1$
(B) $7x$
(C) $7x - 1$
(D) $6x + 1$
(E) $6x$

14. A person receives \$405 for the first month's work and a \$10 a month raise at the beginning of each month thereafter. What will the person's total salary be for the first 6 months?

(A) \$2,760
(B) \$2,580
(C) \$2,480
(D) \$2,440
(E) \$2,430

GO ON TO THE NEXT PAGE



15. In the figure above, the degree measure of the largest of the four angles is how much greater than that of the smallest?

(A) 30 (B) 45 (C) 60 (D) 90 (E) 135

---

16. If $x^2 + 5x + 6 = (x - 1)(x + 6) + r$, then $r =$

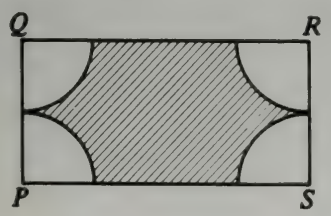
(A) -12 (B) -6 (C) 0 (D) 6 (E) 12

17. In a certain class, on their report cards 10 students received no A's and no B's, 17 students received at least one A, and 13 received at least one B. If 5 students received both A's and B's, how many students are there in the class?

(A) 32
(B) 35
(C) 37
(D) 40
(E) 45

18. On a number line, if point Y is at a distance 3 from point P and point X is at a distance 4 from P, what is the length of XY?

(A) 1 (B) 4 (C) 5 (D) 7
(E) It cannot be determined from the information given.



19. In the rectangle above, the shaded region is bounded in part by 4 equal arcs with centers at P, Q, R, and S. If $PQ = 6$ and $QR = 10$, what is the area of the shaded region?

(A) $60 - 9\pi$
(B) $60 - 4\pi$
(C) $32 - 9\pi$
(D) $32 - 4\pi$
(E) $30 - 6\pi$

20. Today Jack is twice as old as Mary was 10 years ago. How old is Mary today if Jack is now y years old?

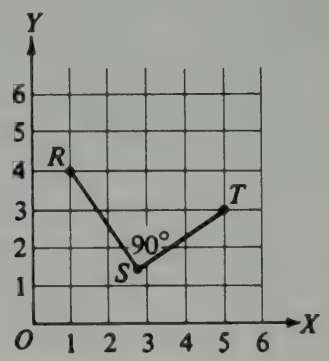
(A) $\frac{y}{2} - 10$

(B) $\frac{y}{2} + 10$

(C) $y + 10$

(D) $2y - 10$

(E) $2(y - 10)$



21. In the graph above, $(RS)^2 + (ST)^2 =$

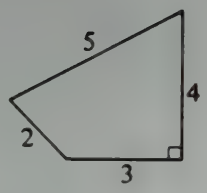
(A) 4 (B) 17 (C) 26 (D) 32 (E) 41

---

22. If $x = \sqrt{18} + \sqrt{32}$, then $x^2 =$

(A) 48
(B) 50
(C) 64
(D) 82
(E) 98

GO ON TO THE NEXT PAGE



23. In the figure above, what is the area of the quadrilateral?

(A) $6 + 2\sqrt{6}$ (B) $2\sqrt{30}$ (C) 11 (D) 13
(E) It cannot be determined from the information given.

24. For how many 3-digit whole numbers is the average value of the digits equal to 1 ?

(A) 1
(B) 3
(C) 4
(D) 5
(E) 6

25. On a map, 1 centimeter represents 4 kilometers. A circle on the map with a circumference of 3π centimeters represents a circular region with what area?

(A) 6π km^2
(B) 12π km^2
(C) 36π km^2
(D) 72π km^2
(E) 144π km^2

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 4

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. LICENSED: (A) unnoticed (B) unwritten (C) unstable (D) not formally authorized (E) not properly trained

2. PREDICAMENT: (A) dignified greeting (B) untroubled situation (C) unexpected increase (D) probability (E) disposition

3. SERENE: (A) persistent (B) unreliable (C) nervous (D) shrewd (E) artificial

4. ARTICULATE:
   (A) state honestly
   (B) affect adversely
   (C) ascribe incorrectly
   (D) speak indistinctly
   (E) admire indiscriminately

5. PERIPHERY:
   (A) inner region
   (B) loose connection
   (C) logical arrangement
   (D) hypothetical relationship
   (E) symmetrical placement

6. OSCILLATING: (A) becoming connected (B) gaining sensation (C) swelling quickly (D) remaining steady (E) opening in stages

7. DISPUTE: (A) dispose (B) answer (C) qualify (D) befriend (E) concede

8. UNWIELDY:
   (A) foreseeable
   (B) vulnerable
   (C) of placid disposition
   (D) of manageable size
   (E) of indeterminate nature

9. ACQUIESCE: (A) disagree (B) depart (C) verify (D) distort (E) persuade

10. CORPULENT: (A) lofty (B) dull (C) spare (D) patient (E) courageous

GO ON TO THE NEXT PAGE

4

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate
(C) sophisticated. .amateur (D) risqué. .crude
(E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. My suggestion was not so casual as it appeared; I ---- mentioned the election because I knew that she wanted to discuss it.

(A) ineptly (B) incessantly (C) inadvertently
(D) intentionally (E) inconsiderately

12. Bureaucratic squabbles can tie up the government for years and ---- domestic and international problems to the point of ----.

(A) magnify. .manageability
(B) reduce. .indignation
(C) intensify. .crisis
(D) weaken. .nonexistence
(E) simplify. .triviality

13. Though there was much ---- among critics over whether or not the play portrayed the "real" Paul Robeson, James Earl Jones's performance was ---- deemed outstanding by audiences and critics.

(A) commentary. .silently
(B) controversy. .unanimously
(C) harassment. .evasively
(D) disappointment. .instructively
(E) discussion. .privately

14. Today science, like philosophy, has been ---- in scope: each has ---- its claim to survey all existence and has accepted a province which, if great, is yet smaller than the whole.

(A) limited. .reiterated (B) narrowed. .resigned
(C) expanded. .repudiated (D) fluctuating. .denied
(E) receding. .amplified

15. Extremist advocates of the occult claim that existing systems of scientific thought must at least be modified, if not ----.

(A) evaluated (B) supplemented
(C) incorporated (D) implemented
(E) overturned

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. VEIN : BLOOD :: (A) wire : cord
(B) bone : membrane (C) nose : scent
(D) pipe : water (E) tunnel : car

17. REFEREE : CONTEST :: (A) judge : trial
(B) defendant : lawsuit (C) noncombatant : war
(D) member : union (E) spectator : game

18. SYNONYM : THESAURUS ::
(A) date : card catalog (B) definition : dictionary
(C) list : roster (D) quotation : novel
(E) script : playbill

19. CLASP : HAND :: (A) shoe : foot
(B) blink : eye (C) embrace : arm
(D) crutch : leg (E) shake : head

20. FIDGET : RESTLESS :: (A) beckon : busy
(B) quibble : content (C) ponder : lazy
(D) obey : impudent (E) bustle : hurried

21. HOMING PIGEON : BIRD :: (A) balloon : rocket
(B) boomerang : missile (C) dirigible : kite
(D) fledgling : sparrow (E) bullet : revolver

22. INKLING : CERTITUDE ::
(A) portent : omen
(B) hope : despair
(C) skepticism : dubiety
(D) misgiving : dread
(E) pomposity : courtesy

23. FLORID : EMBELLISHMENTS :: (A) candid : ideas
(B) verbose : words (C) blatant : voices
(D) chimerical : facts (E) monotonous : rhythms

24. HOMELESS : DOMICILE :: (A) childless : orphan
(B) impregnable : fortress (C) thirsty : hunger
(D) insolvent : money (E) alien : journey

25. DICTUM : AUTHORITATIVE ::
(A) libel : defamatory (B) proverb : ethical
(C) oath : laudatory (D) accusation : punitive
(E) tenet : educational

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Crete was settled about 5000 B.C. by migrants from Greece who journeyed across 60 miles of open sea to reach this island in the Mediterranean. From the beginning the land was too poor for cultivation. To survive, the people had to take advantage of their geographic location. Situated halfway between Egypt and Greece, the Cretans were the first people to develop a modern commercial economy based predominantly on seafaring and trade.

Using evidence from various sources, we can reconstruct some of the details of Cretan life. Archaeological digs support the legend that the people of Crete were ingenious artisans. We know, for example, that shipbuilding was a high point in their technology. The island was well forested, providing timber for ships. Their vessels were seaworthy and must have required careful preliminary designs and expert joinery. They had high, sharply upcurved bows and sterns, and keels plated in bronze that projected some distance beyond the prow, a design which suggests that the boats were intended for ramming. This naval tactic, developed in Crete, dominated sea warfare in classical times.

Because of the inventiveness of the people of Crete, their communities were more prosperous than those of other early civilizations. Trading left the populace with leisure time; bullfights are known to have been a popular diversion. Frescoes show figures dressed in colorful clothes, large gaily decorated hats, and elaborate jewelry. The remains of open-patterned villages suggest an essentially middle-class society. The palace of the priest-kings, the Minos, was as much a commercial center as a palace, reflecting the importance of trade in the Cretan culture.

26. The primary purpose of the passage is to
    (A) describe an early culture
    (B) explain the technology of shipbuilding
    (C) comment on the origin of the settlers of Crete
    (D) analyze the advantages of a trade-oriented economy
    (E) speculate about the relationship between geography and culture

27. According to the passage, all of the following statements about Crete are true EXCEPT:
    (A) The land was not especially suitable for the development of an agricultural economy.
    (B) The location of the island enabled the settlers to make a good living by trade.
    (C) The island had a well-developed civilization before migrants from Greece arrived.
    (D) The island provided an abundant supply of timber.
    (E) The island was about as far from Greece as it was from Egypt.

28. The tone of the passage can best be described as
    (A) whimsical (B) argumentative
    (C) contradictory (D) impatient
    (E) objective

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4

Mr. Casaubon spent a great deal of time at the Grange during the weeks of his courtship, and the hindrance which this occasioned to the progress of his life's work—*The Key to All Mythologies*—naturally made him look forward the more eagerly to the happy termination of courtship. But he had deliberately incurred the hindrance, having made up his mind that it was now time to adorn his life with the graces of female companionship, to irradiate the gloom which fatigue was apt to cause with the play of female fancy, and to secure in this, his culminating age, the solace of female tendance for his declining years. Hence he abandoned himself to the stream of feeling, and was surprised to find what an exceedingly shallow rill it was. As in droughty regions baptism by immersion could only be performed symbolically, so Mr. Casaubon found that sprinkling was the utmost approach to a plunge which his stream would afford him; and he concluded that the poets had much exaggerated the force of masculine passion. Nevertheless, he observed with pleasure that Miss Brooke showed a submissive affection which promised to fulfill his most agreeable provisions for marriage. It had crossed his mind that possibly there was some deficiency in Dorothea to account for the moderation of his abandonment, but he was unable to discern the deficiency or to figure to himself a woman who would have pleased him better, so that there was clearly no reason to fall upon but the exaggeration of human tradition.

29. Which of the following best expresses Casaubon's principal reason for wanting to be married?

(A) He is passionately in love with Dorothea.
(B) He desires present cheering and future comforting.
(C) He wants to make both Dorothea and himself happy.
(D) He thinks marriage will improve the quality of his scholarship.
(E) He feels it is time to regulate his life and raise a family.

30. It can be inferred that Casaubon views courtship primarily as

(A) an irksome sacrifice to Dorothea's whims
(B) a joy that one should experience before old age sets in
(C) an essential part of the symbolism of marriage
(D) a tradition based on foolish prohibitions
(E) a convention to be observed

31. Which of the following best sums up Casaubon's reasoning about his dissatisfaction with courtship?

(A) Dorothea is deficient in affection; hence, I am disappointed.
(B) I am no longer young; therefore, I must try to moderate my passions.
(C) I simply do not have the temperament for courtship; it can therefore only be disappointing.
(D) People overstate the pleasures of courtship; it cannot live up to its reputation.
(E) Courtship does not take strong emotions into account; it is an outdated tradition.

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During the 1960's and 1970's, brain researchers developed the very useful, though provisional, "amine hypothesis": depression results from a chemical imbalance of neurotransmitters in the brain. Neurotransmitters are chemicals that carry messages from one nerve cell to the next at lightning speed, telling it to fire or resist firing. Over a dozen neurotransmitters have been identified, but the three most closely studied with respect to mental illness are norepinephrine, dopamine, and serotonin, amines derived from essential amino acids.

The first clue to the role of these three amines came in the late 1950's, when physicians noted that patients taking the drug reserpine to control hypertension became severely depressed, while tubercular patients treated with a new drug, iproniazid, suddenly felt elated. Biochemists studying the brains of rats then discovered that reserpine depleted their stores of norepinephrine and serotonin, whereas iproniazid had the opposite effect.

The amine hypothesis was temporarily threatened by the discovery of the tricyclic antidepressants, new compounds that proved extremely effective against depression but that did not increase the total level of amines in the brains of laboratory animals. The mystery was resolved a few years later when Dr. Julius Axelrod described the way in which norepinephrine is turned off after it has relayed its message at a nerve synapse (the junction between nerve cells): it is reabsorbed by the nerve endings from which it came, to be conserved for future use. Axelrod showed that the tricyclics interfere with this process, thus increasing the amount of norepinephrine that is available to receptors on the next nerve cell. The same proved true of serotonin, and the amine hypothesis was saved.

32. The author's general purpose is to

(A) illustrate the complexity of nerve cell structure
(B) consider competing theories of mental illness
(C) present a biochemical explanation of depression
(D) evaluate the effectiveness of antidepressant drugs
(E) suggest future possibilities for brain research

33. Which of the following can be inferred from the second paragraph?

(A) The combination of reserpine and iproniazid is effective against depression.
(B) Depression is associated with a deficit of norepinephrine and serotonin.
(C) Laboratory animals are more sensitive to reserpine and iproniazid than are humans.
(D) Certain neurotransmitters are themselves causes of depression.
(E) The effects of reserpine and iproniazid vary with brain size.

34. According to the passage, the tricyclics operate by

(A) obstructing the synthesis of amines
(B) freeing certain neurotransmitters for immediate utilization
(C) acting as substitutes for norepinephrine
(D) reducing the nerve cells' need for neurotransmitters
(E) enabling nonessential amines to be used as neurotransmitters

35. Which of the following statements about the amine hypothesis is LEAST supported by the passage?

(A) It is useful in curing a wide variety of brain malfunctions.
(B) It was suggested by seemingly unrelated medical practices.
(C) It may be modified after further experimentation.
(D) It has on occasion been called into question.
(E) It proposes that certain mental disorders are chemically caused.

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4

Literary critics have had their go at humor, and I have read some of their interpretations of humorous literature, but without being greatly instructed. Humor can be dissected, as a frog can, but the thing dies in the process and the innards are discouraging to any but the purely scientific mind. Humor has a certain fragility, an evasiveness, which one had best respect. Essentially, it is a mystery.

One of the things commonly said about humorists is that they are really very sad people—clowns with broken hearts. There is some truth in this, but it would be more accurate, I think, to say that there is a deep vein of melancholy running through everyone's life and that a humorist, perhaps more sensible of it than some others, compensates for it actively and positively. Practically everyone is a manic depressive of sorts, with up moments and down moments, and you certainly do not have to be a humorist to taste the sadness of situation and mood. But there is often a rather fine line between laughing and crying, and if a humorous piece of writing brings people to the point where their emotional responses are untrustworthy and seem likely to break over into the opposite realm, it is because humorous writing, like poetical writing, has an extra dimension. It plays, like an active child, close to the big hot fire which is Truth. And sometimes the reader feels the heat.

The world likes humor but treats it patronizingly. It decorates serious artists with laurel and wags with brussels sprouts. It feels that if a thing is funny it is something less than great, because if it were truly great it would be wholly serious. Writers know this, and those who take their literary selves with great seriousness are at considerable pains never to associate their names with anything funny or flippant or nonsensical or "light." They suspect it would hurt their reputations, and they are right. Many poets today sign their real name to their serious verse and a pseudonym to their comical verse, being unwilling to have the public discover them in any but a pensive and sober moment.

Many humorists, in the early stages of their careers, have been asked anxiously by friends when they were "going to write something serious." That occasion is memorable, for it gives them pause to realize that the bright star they are following is held to be not of the first magnitude.

36. The author states that melancholy is a condition

(A) conducive to egotism
(B) universal to humanity
(C) inimical to mystery
(D) caused by insecurity
(E) indicative of sensitivity

37. The author contends that humorists lack

(A) esteem (B) insight
(C) emotion (D) popularity (E) subtlety

38. According to the passage, the scientific analysis of humor is

(A) dull but necessary
(B) detailed but obvious
(C) commonplace but misunderstood
(D) possible but inadequate
(E) impressive but difficult

39. Which of the following would the author probably consider most favorable to the future of humor?

(A) More detailed analyses of humorous literature
(B) More frequent use of pseudonyms by humorists
(C) More open-mindedness on the part of the reading public
(D) The curtailment of awards for literary works
(E) The development of a method for treating the manic-depressive personality

40. It can be inferred that the "bright star" mentioned in line 47 of the passage stands for which of the following?

(A) Popularity (B) Wisdom (C) Superiority
(D) Wealth (E) Humor

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$: Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

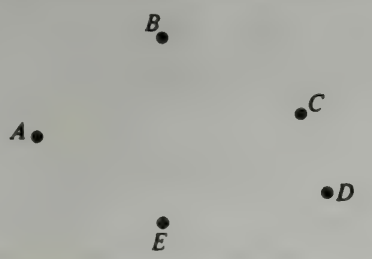
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1. Which of the following can be divided both by 2 and by 3 with no remainders?

   (A) 56 (B) 63 (C) 76 (D) 81 (E) 96

2. If a synthetic rubber factory can produce 75 pounds of rubber every 30 seconds, how many pounds of rubber can this factory produce per hour?

   (A) 150 (B) 900 (C) 9,000
   (D) 15,000 (E) 135,000



3. How many different line segments can be drawn to connect all possible pairs of points shown above?

   (A) 32 (B) 25 (C) 20 (D) 10 (E) 5

Candy	Cost
A	2 pieces for 5 cents
B	3 pieces for 4 cents
C	4 pieces for 7 cents
D	5 pieces for 6 cents
E	6 pieces for 9 cents

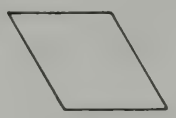
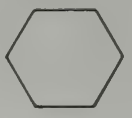
4. According to the table above, which candy costs the least per piece?

   (A) A (B) B (C) C (D) D (E) E

5. Pat rode 5 miles each way from home to the beach at an average rate of 10 miles per hour. If she spent exactly 6 hours at the beach and made no other stops, how many hours was she gone from home?

   (A) $6\frac{1}{2}$ (B) 7 (C) $7\frac{1}{2}$ (D) 8 (E) 10

6. A figure in the plane is said to be "circle-receptive" if a circle touching every side of the figure can be drawn inside the figure, and a circle touching every vertex of the figure can be drawn outside the figure. Which of the following figures is NOT "circle-receptive"?

   (A) 
   (B) 
   (C) 
   (D) 
   (E) 

7. If a shirt is priced at \$8.50, how much will it cost when its price is reduced by 10 per cent?

   (A) \$0.85
   (B) \$7.65
   (C) \$7.75
   (D) \$8.40
   (E) \$8.65

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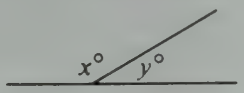
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

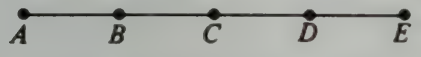
A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	The number of minutes in a day	The number of days in a year
9.	x if $2x - 14 = 0$	y if $y - 7 = 0$
10.	The greatest prime number between 15 and 20	17



Lettered points are equally spaced.

11.	BD	$\frac{AE}{2}$

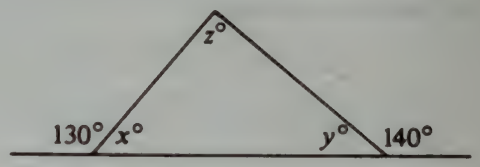
If half of N is added to twice N, the sum is 90.

12.	N	32
13.	740×0.2	$\frac{1}{4}$ of 740

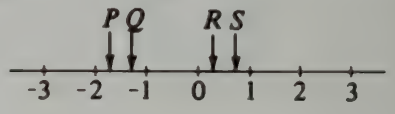
	Column A	Column B

In $\triangle PQR$, the sum of the degree measures of $\angle Q$ and $\angle R$ is greater than $90°$.

14.	Degree measure of $\angle P$	Degree measure of $\angle Q$
15.	$56(842) + 44(842)$	84201



16.	$x + y$	z
17.	x if $11 < x < 17$	y if $12 < 2y < 20$



Number Line

18.	$P \times Q$	$R \times S$
19.	$(4^3)^2$	$4^{(3^2)}$
20.	The difference between two numbers each of which is between 2 and 5	The sum of two numbers each of which is between 1 and 2

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Column A | Column B

Circle R has diameter 6.
Circle S has circumference 4π.

21. Area of R | Area of S

$pq > 0$
$pr < 0$

22. qr | 0

23. The volume of a cylinder with height 5 | The volume of a cone with height 5

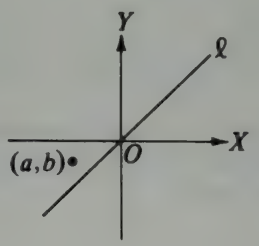
$x > y > 0$

24. (20% of x) + (30% of y) | 50% of $(x + y)$

a and b are positive.

25. $\frac{a}{b}$ | $\frac{a+1}{b+1}$

Column A | Column B



Line ℓ has the equation $y = x$.

26. a | b

S is the set of all integers greater than 0 and less than 100.

27. Number of integers N in S such that both $\frac{N}{2}$ and $\frac{N}{5}$ are in S | Number of integers K in S such that $\frac{K}{10}$ is in S

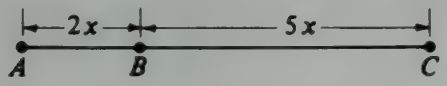
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5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

28. If the present time is 7 a.m., what time was it 40 hours ago?

(A) 3 a.m.
(B) 11 a.m.
(C) 4 p.m.
(D) 3 p.m.
(E) 11 p.m.



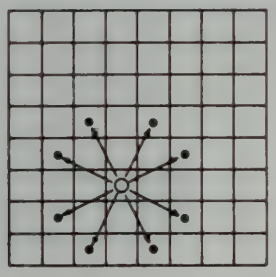
29. If segment AC above has length 35, then segment BC is how much longer than segment AB ?

(A) 7
(B) 10
(C) 15
(D) 21
(E) 25

---

30. If $x + y = 5$ and $x + 2y = 10$, then $x - y =$

(A) -5 (B) -4 (C) 0 (D) 4 (E) 5



31. In a certain game, a piece is allowed to move as shown by the arrows above. On an empty board, what is the number of spaces from which all 8 such moves are NOT possible?

(A) 16 (B) 24 (C) 32 (D) 48 (E) 56

---

32. If the sum of three numbers is 3 more than the average of these three numbers, then the sum of these numbers is

(A) $-\frac{9}{2}$ (B) 3 (C) $\frac{9}{2}$ (D) 6 (E) 9



33. In the figure above, five semicircles are drawn on the number line. What is the total shaded area?

(A) $\frac{5\pi}{16}$

(B) $\frac{5\pi}{8}$

(C) $\frac{5\pi}{4}$

(D) $\frac{5\pi}{2}$

(E) 5π

---

34. The fare schedule for a taxi ride is 75 cents for the first quarter of a mile and 15 cents for each additional quarter of a mile. What is the fare in cents for a ride of x miles, where x is an integer?

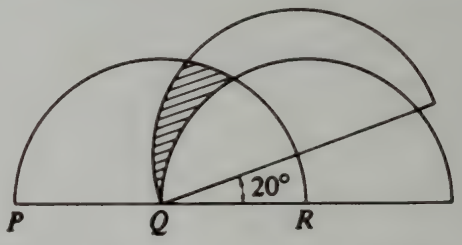
(A) $75 + 15x$

(B) $75 + 45x$

(C) $75 + 15(4x)$

(D) $75 + 15\left(x - \frac{1}{4}\right)$

(E) $75 + 15(4x - 1)$



35. If the area of each of the three semicircles shown above is 72 and Q is the midpoint of segment PR, what is the area of the shaded region?

(A) 4 (B) 8 (C) 9 (D) 12 (E) 20

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test Form Code 4X			
VERBAL		MATHEMATICAL	
Section 1.	Section 4.	Section 2.	Section 5.
1. C	1. D	1. C	1. E
2. C	2. B	2. C	2. C
3. D	3. C	3. E	3. D
4. E	4. D	4. A	4. D
5. B	5. A	5. D	5. B
6. E	6. D	6. A	6. A
7. A	7. E	7. B	7. B
8. C	8. D	8. E	*8. A
9. B	9. A	9. D	*9. C
10. D	10. C	10. C	*10. A
11. C	11. D	11. C	*11. C
12. E	12. C	12. D	*12. A
13. E	13. B	13. D	*13. B
14. E	14. B	14. B	*14. D
15. C	15. E	15. D	*15. B
16. B	16. D	16. E	*16. C
17. A	17. A	17. B	*17. A
18. D	18. B	18. E	*18. A
19. E	19. C	19. A	*19. B
20. E	20. E	20. B	*20. D
21. C	21. B	21. B	*21. A
22. E	22. D	22. E	*22. B
23. B	23. B	23. A	*23. D
24. B	24. D	24. E	*24. B
25. D	25. A	25. C	*25. D
26. B	26. A		*26. B
27. A	27. C		*27. C
28. D	28. E		28. D
29. C	29. B		29. C
30. B	30. E		30. A
31. B	31. D		31. D
32. B	32. C		32. C
33. E	33. B		33. B
34. E	34. B		34. E
35. A	35. A		35. B
36. D	36. B		
37. B	37. A		
38. E	38. D		
39. C	39. C		
40. A	40. E		
41. A			
42. A			
43. B			
44. E			
45. C			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 304. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 304. Enter this figure on the worksheet.

FORM CODE 4X

SAT SCORING WORKSHEET

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 4X

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	450	600
84	780		39	450	590
83	770		38	440	590
82	760		37	430	580
81	750		36	430	570
80	740		35	420	560
79	730		34	410	550
78	720		33	410	540
77	710		32	400	530
76	700		31	390	520
75	690		30	390	520
74	680		29	380	510
73	670		28	370	500
72	670		27	370	490
71	660		26	360	480
70	650		25	350	470
69	650		24	350	460
68	640		23	340	450
67	630		22	330	440
66	630		21	330	440
65	620		20	320	430
64	610		19	310	420
63	610		18	310	410
62	600		17	300	400
61	590		16	290	390
60	590	800	15	290	380
59	580	790	14	280	370
58	570	780	13	270	370
57	570	770	12	270	360
56	560	760	11	260	350
55	550	750	10	250	340
54	550	740	9	250	330
53	540	730	8	240	320
52	530	720	7	230	310
51	530	710	6	230	300
50	520	700	5	220	290
49	510	690	4	210	290
48	510	680	3	210	280
47	500	670	2	200	270
46	490	660	1	200	260
45	490	650	0	200	250
44	480	640	–1	200	240
43	470	630	–2	200	230
42	470	620	–3	200	220
41	460	610	–4	200	220
			–5	200	210
			–6	200	200
			or below		